Fact File

Cloud Seeding or Artificial Rainfall

People have always wanted to create rain, so that they would not suffer from drought. Modern science has been successful in causing rain in a limited way through cloud seeding. This method is based on the knowledge of growing ice crystals in clouds.

One method to cause rainfall from clouds is to introduce particles of dry ice (solid CO_2) into the cloud from an air plane. The dry ice causes ice crystals to form in the cloud. These ice crystals coalesce, grow, melt and fall as rain. Cloud seeding will not be successful unless the cloud is already saturated with water vapour.

by the National Hurricane Centre. They are now maintained and updated by an international committee of the World Meteorological Organization (WMO).

Large scale destruction caused by Odisha cyclone in 1999, triggered the issue of naming tropical cyclones developed in the North Indian ocean. As a result, naming conventions for storms that develop in the Indian Ocean began in 2004. WMO (World Meteorological Organisation)had informed each of the eight South Asian member countries to submit a list of their own eight names for the cyclones.

DO YOU KNOW?

An isohyets or isohyetal **line** is a **line** joining points of **equal rainfall** on a map in a given period. A map with isohyets is called an isohyetal map.

6.7.3 Tornado and Water Spouts

It is a very small intense, funnel shaped very speed whirl wind system. Its speed and direction of the movement are erratic (Figure 6.33). The winds are always as fast as 500 km ph. The fast moving air converges in the middle and rises up. The uplift is capable of rising dust, trees and other weaker objects in its path. South and western part of Gulf States of USA experiences frequent tornados.

Water spouts are formed over water body similar to tornados in the formation and structure. This sometimes leads to fish rain, if the mass of fish comes under the water spout.

Web link for Water spout at Chennai, Tamil Nadu https://www.youtube.com/watch?v=v0RubwHxlgM



Figure 6.33 Tornado

B. Temperate Cyclone

The cyclone formed in the mid latitudes is called as temperate cyclone. As they are formed due to movement of air masses and front, they are called as 'Dynamic cyclone' and 'Wave cyclone'. This cyclone is characterised by the four different sectors, which are varied with their weather patterns (Figure 6.34).

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